YEAR 10 PHYSICS, AQA GCSE, PART 2 OF 3

PARTICLE MODEL OF MATTER

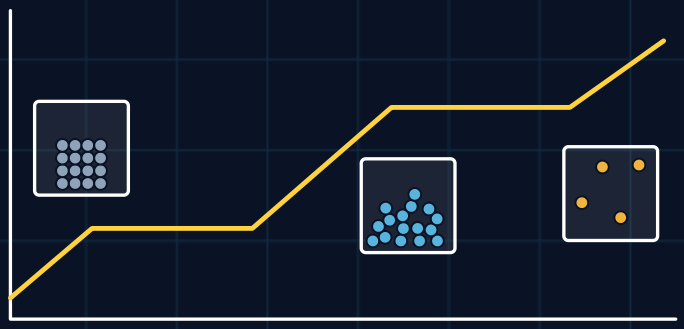
PART 2 OF 3

LESSONS 3 AND 5

IN THIS BOOKLET

- Lesson 3: Density
- Lesson 5: Internal energy and heating curves

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lessons 3 and 5** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try “Your turn” alone.

TRY IT
Practice that builds up, then a challenge. Every second lesson ends with the slides' Quick check.

PRACTICAL
Record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS BOOKLET HIGHLIGHTED)

01	Solids, liquids and gases	02	Gas particles and pressure
03	**Density** Slides 13–17	04	Measuring density (required practical)
05	**Internal energy and heating curves** Slides 24–28	06	Specific latent heat
07	Specific heat capacity	08	Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

03 DENSITY

AQA Physics 4.3.1.1; Combined Science 6.3.1.1

BIG QUESTION **A gold bar the size of a carton of milk is too heavy for most people to lift with one hand. Why?**

BY THE END YOU CAN

- calculate density with $\rho = \frac{m}{V}$
- use the units kg/m³ and g/cm³
- explain differences in density using the particle model
- rearrange the equation to find a mass or a volume

density **mass** **volume** **kilogram per cubic metre (kg/m³)** **rho (ρ)**

YOU WILL NEED

Per pair: calculator. Demonstration: blocks of the same size made of different materials (wood, aluminium, steel) to pass round.

Teaching deck, slides 13–17
panphy.app/gcsephy/decks/particle-model/#13

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Describe the arrangement of the particles in a solid.

Very close together, in a regular pattern.

2
A gas in a sealed can is heated. What happens to its pressure?

It increases.

3
Convert 250 g to kilograms.

0.250 kg

EXPLAIN

HOW MUCH MASS IS PACKED IN?

Pick up a block of steel and a block of wood of the same size. The steel has much more mass, because more mass is packed into the same volume. We say that steel has a greater **density**.

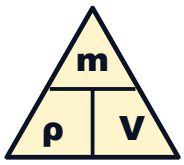
! **Density** is the mass of each unit of volume: the mass of 1 m³ (or of 1 cm³) of the material.

$$\rho = \frac{m}{V}$$

density = mass ÷ volume

ρ in kg/m³, m in kg, V in m³

Rearranged: $m = \rho V$ and $V = \frac{m}{\rho}$. The symbol ρ is the Greek letter rho.



WITH YOUR TEACHER SLIDE 14

A block of aluminium has a mass of 5400 kg and a volume of 2.0 m³. Calculate its density.

EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{5400}{2.0}$
ANSWER	$\rho = 2700$ kg/m³

YOUR TURN

A tank holds 3.0 m³ of water. The mass of the water is 3000 kg. Calculate the density of water.

EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{3000}{3.0}$
ANSWER	$\rho = 1000$ kg/m³

TWO SETS OF UNITS

In the lab it is easier to measure mass in grams and volume in cubic centimetres. Then the density is in **g/cm³**. Use one set of units or the other; never mix them.

MATERIAL	DENSITY IN G/CM³	DENSITY IN KG/M³
air	0.0012	1.2
ice	0.92	920
water	1.0	1000
aluminium	2.7	2700
iron	7.9	7900
gold	19.3	19 300

> **Get the marks:** To change g/cm³ into kg/m³, **multiply by 1000**. To change kg/m³ into g/cm³, divide by 1000. Water is the one to remember: 1.0 g/cm³ = 1000 kg/m³.

DENSITY AND THE PARTICLE MODEL

In a solid and in a liquid the particles are close together, so a lot of mass fits into each cubic centimetre: solids and liquids have high densities. In a gas the particles are far apart, with empty space between them, so a gas has a very low density. Water is about 800 times denser than air.

Most substances are a little denser as a solid than as a liquid, because the particles are packed slightly closer. Water is unusual: ice is *less* dense than liquid water, which is why ice floats.

FINDING THE MASS OR THE VOLUME

WITH YOUR TEACHER SLIDE 17

The density of iron is 7900 kg/m³. Calculate the mass of 0.020 m³ of iron.

EQUATION $\rho = \frac{m}{V}$

REARRANGE $m = \rho V$

SUBSTITUTE $m = 7900 \times 0.020$

ANSWER **m = 158 kg**

YOUR TURN

The density of aluminium is 2700 kg/m³. Calculate the mass of 0.050 m³ of aluminium.

EQUATION $\rho = \frac{m}{V}$

REARRANGE $m = \rho V$

SUBSTITUTE $m = 2700 \times 0.050$

ANSWER **m = 135 kg**

WITH YOUR TEACHER SLIDE 17

The density of gold is 19.3 g/cm³. Calculate the volume of a 965 g gold bar.

EQUATION $\rho = \frac{m}{V}$

REARRANGE $V = \frac{m}{\rho}$

SUBSTITUTE $V = \frac{965}{19.3}$

ANSWER **V = 50 cm³**

YOUR TURN

The density of aluminium is 2.7 g/cm³. Calculate the volume of a 540 g aluminium block.

EQUATION $\rho = \frac{m}{V}$

REARRANGE $V = \frac{m}{\rho}$

SUBSTITUTE $V = \frac{540}{2.7}$

ANSWER **V = 200 cm³**

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The unit of density is | kg | **kg/m³** | m³/kg |

A gas has a low density because its particles are

| very light | **far apart** | moving quickly |

2 SUPPORTED A stone has a mass of 240 g and a volume of 80 cm³. Calculate its density in g/cm³.

EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{240}{80}$
ANSWER	**3.0 g/cm³** unit:

3 SUPPORTED A concrete beam has a mass of 4800 kg and a volume of 2.0 m³. Calculate the density of the concrete.

EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{4800}{2.0}$
ANSWER	**2400 kg/m³** unit:

4 Complete the table. Show your working underneath.

Object	Mass	Volume	Density
Wooden block	360 g	600 cm³	0.60 g/cm³
Copper pipe	445 g	50 cm³	8.9 g/cm³
Air in a room	60 kg	50 m³	1.2 kg/m³
Petrol in a tank	36 kg	0.050 m³	720 kg/m³

5 The density of sea water is 1.03 g/cm³. Write this density in kg/m³.

1.03 × 1000 = 1030 kg/m³

6 Explain, using the particle model, why 1 cm³ of liquid water has a much greater mass than 1 cm³ of steam.

In liquid water the particles are close together, but in steam they are far apart. So 1 cm³ of water contains many more particles, and more mass, than 1 cm³ of steam.

7 CHALLENGE A crown has a mass of 1.9 kg and a volume of 120 cm³. The density of gold is 19.3 g/cm³. Is the crown pure gold? Show your working. *Convert the mass to grams first.*

m = 1900 g. $\rho = \frac{m}{V} = \frac{1900}{120} = 15.8$ g/cm³. This is less than 19.3 g/cm³, so the crown is not pure gold.

AQA-STYLE EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** Write the equation in symbols first, then the numbers. A correct equation often earns a mark even if the arithmetic goes wrong.

8 A block of ice has a mass of 2760 kg and a volume of 3.0 m³. Calculate the density of ice. **[2 marks]**

EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{2760}{3.0}$ (1)
ANSWER	**920 kg/m³ (1)** unit:

9 A fuel tank holds 0.060 m³ of diesel. The density of diesel is 850 kg/m³. Calculate the mass of the diesel. **[3 marks]**

EQUATION	$\rho = \frac{m}{V}$
REARRANGE	$m = \rho V$ (1)
SUBSTITUTE	$m = 850 \times 0.060$ (1)
ANSWER	**51 kg (1)** unit:

10 Explain why a gas has a much lower density than a solid. **[2 marks]**

The particles in a gas are much further apart than in a solid (1), so there is less mass (fewer particles) in the same volume (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Density is the mass per unit volume: $\rho = \frac{m}{V}$.
- Units: kg/m³, or g/cm³ in the lab. 1 g/cm³ = 1000 kg/m³.
- Rearranged: $m = \rho V$ and $V = \frac{m}{\rho}$.
- Solids and liquids are dense because their particles are close together. Gases have low densities because their particles are far apart.

COVER THE PAGE AND ANSWER

1 Write the equation for density.

2 State the unit of density.

3 What is the density of water in kg/m³?

4 A block has a mass of 80 g and a volume of 20 cm³. What is its density?

I CAN...

- calculate density with $\rho = \frac{m}{V}$
- use kg/m³ and g/cm³ and convert between them
- rearrange the equation to find a mass or a volume
- explain differences in density using the particle model

Lesson

05 INTERNAL ENERGY AND HEATING CURVES

AQA Physics 4.3.2.1, 4.3.2.3; Combined Science 6.3.2.1, 6.3.2.3

BIG QUESTION **A pan of water boils on the hob for ten minutes. The hob keeps supplying energy, but the water never gets hotter than 100 °C. Where does the energy go?**

BY THE END YOU CAN

- state what internal energy is
- describe the two things that can happen when a substance is heated
- explain the flat sections of a heating or cooling graph
- read melting and boiling points from a graph

internal energy **kinetic energy** **potential energy** **melting point** **boiling point** **heating curve**

YOU WILL NEED

Per group: boiling tube of molten stearic acid (from a hot water bath), thermometer, clamp stand, stopwatch. Or use the sample results.

Teaching deck, slides 24–28
panphy.app/gcsephy/decks/particle-model/#24

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Write the equation for density.

$\rho = \frac{m}{V}$

2
What is the temperature of a gas related to?

The average kinetic energy of its molecules.

3
Name the change of state from liquid to solid.

Freezing.

EXPLAIN

THE ENERGY INSIDE

The particles of a substance store energy in two ways. They are moving (or vibrating), so they have **kinetic energy**. They also have **potential energy**, which depends on how far apart they are: energy has to be supplied to pull particles away from each other.

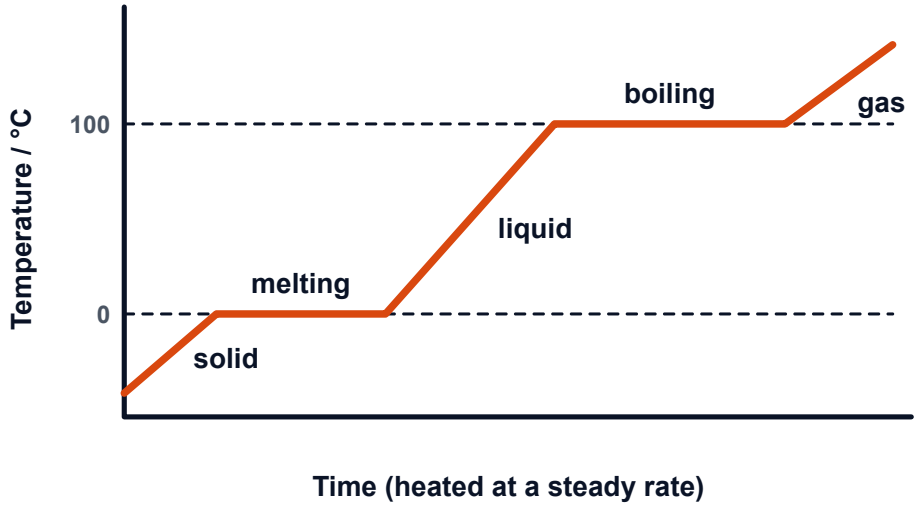
! **Internal energy** is the total kinetic energy and potential energy of all the particles (atoms and molecules) that make up a system.

HEATING DOES ONE OF TWO THINGS

Heating a substance increases the energy of its particles, so its internal energy goes up. This **either** raises the temperature **or** changes the state. It does not do both at once.

	THE TEMPERATURE RISES	THE STATE CHANGES
What the particles do	move or vibrate faster	break away from their neighbours and move further apart
Which energy increases	kinetic energy	potential energy
Temperature	increases	stays constant

A HEATING GRAPH



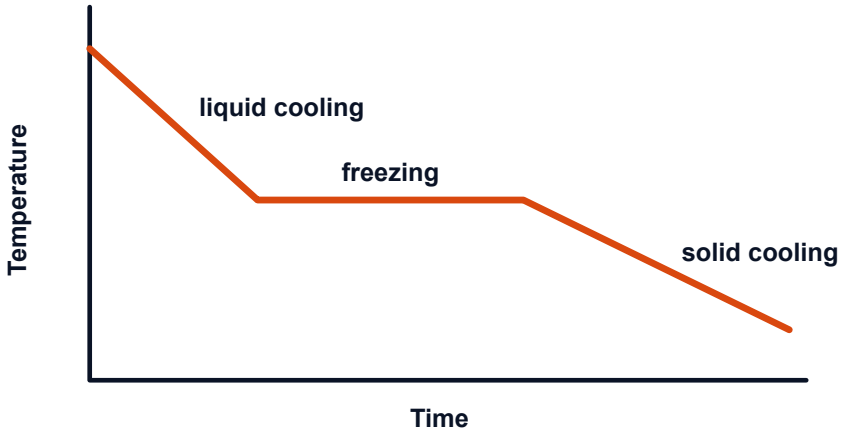
Ice from the freezer, heated at a steady rate until it is steam

- **Sloping sections:** the substance is in one state and is getting hotter. The kinetic energy of the particles is increasing.
- **Flat sections:** the substance is changing state. The energy supplied increases the potential energy of the particles, so the temperature does not change.
- The first flat section is at the **melting point** (0 °C for water). The second is at the **boiling point** (100 °C for water).

> **Get the marks:** On a flat section, heating has **not** stopped. Energy is still being supplied. It is being used to change the state, not to raise the temperature.

A COOLING GRAPH

Cooling is the reverse. As a liquid cools, its particles slow down and the temperature falls. While it **freezes**, the particles move closer into fixed positions and their potential energy decreases: energy is transferred to the surroundings, but the temperature stays constant. Then the solid cools.



PRACTISE

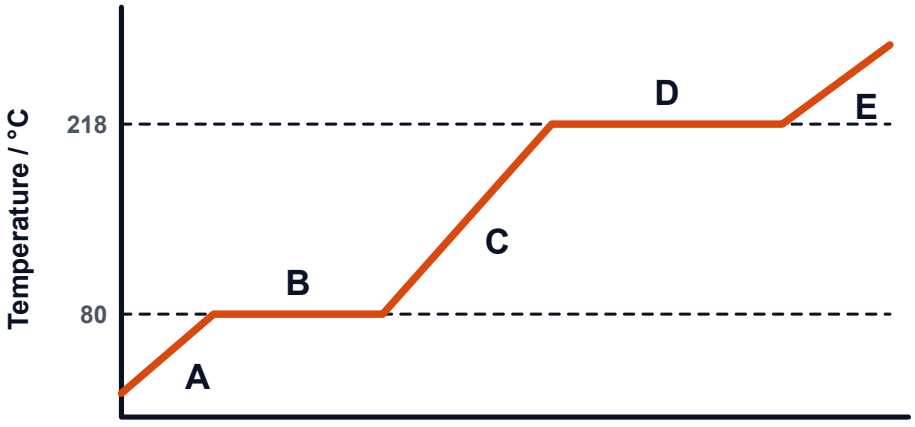
TRY IT

1 Complete the sentences.

Internal energy is the total ___kinetic___ **(kinetic / thermal / elastic)** energy and ___potential___ **(chemical / potential / nuclear)** energy of all the particles in a system.

While a substance is melting, its temperature ___stays the same___ **(increases / decreases / stays the same)**.

2 A solid is heated at a steady rate until it has become a gas. The graph shows how its temperature changes.



(a) Which section shows the substance melting?

A	**B**	C	D	E

(b) Which section shows the liquid getting hotter? | A | B | **C** | D | E |

(c) State the melting point and the boiling point of the substance.

Melting point 80 °C; boiling point 218 °C.

(d) In which state is the substance at 150 °C?

Liquid.

3 Draw a ring around the correct answer in each box.

When water at 50 °C is heated to 60 °C, its particles gain

kinetic energy	potential energy	no energy

When water boils at 100 °C, its particles gain

kinetic energy	**potential energy**	no energy

4 **CHALLENGE** 1 kg of steam at 100 °C and 1 kg of water at 100 °C are at the same temperature. Explain which has the greater internal energy.

The steam. The particles have the same average kinetic energy (same temperature), but the particles in the steam are much further apart, so they have more potential energy.

INVESTIGATE

PRACTICAL: A COOLING CURVE

Hot liquid: wear eye protection and do not touch the tube.

STAY SAFE

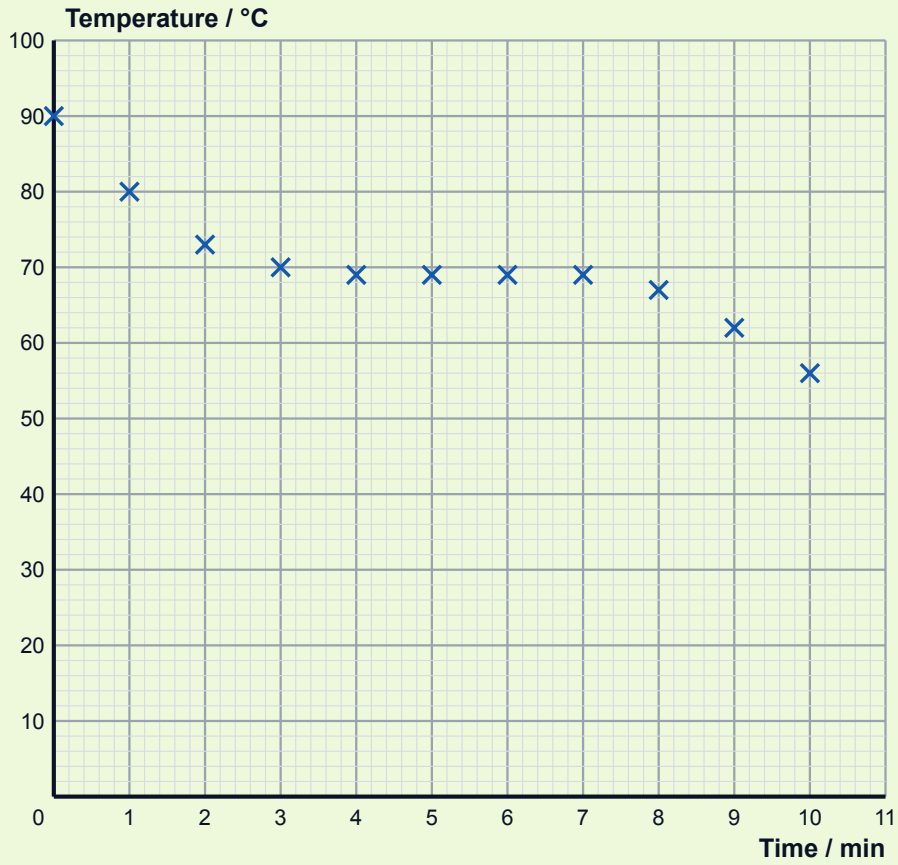
- Wear eye protection.
- The tube and the water bath are hot: leave the tube in its clamp.
- Do not stir with the thermometer once the acid starts to set: it may break.

A tube of melted stearic acid (a waxy solid at room temperature) is left to cool in air. Record its temperature every minute. If you are not doing the practical, use the sample results your teacher gives you.

5 Record the temperature.

Time / min	0	1	2	3	4	5	6	7	8	9	10
Temperature / °C	90	80	73	70	69	69	69	69	67	62	56

6 Plot a graph of temperature against time. Draw a smooth line through your points.



7 Use your graph to find the freezing point of stearic acid.

About 69 °C: the temperature of the flat section.

8 Explain why the temperature stays the same while the stearic acid freezes, even though it is still transferring energy to the air.

The particles are moving closer together into fixed positions, so their potential energy is decreasing. Their kinetic energy does not change, so the temperature stays constant.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: For a flat section, name the change of state, say that energy is still being supplied, and say which energy of the particles changes.

9 A block of ice at 0 °C is heated. Its temperature does not change until all the ice has melted. Explain why. **[3 marks]**

The energy supplied is used to change the state (melt the ice) (1). It increases the potential energy of the particles (as they break away from their fixed positions) (1), not their kinetic energy, so the temperature stays the same (1).

10 State what is meant by the internal energy of a system. **[2 marks]**

The total kinetic energy and potential energy (1) of all the particles that make up the system (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Internal energy = total kinetic + potential energy of all the particles.
- Heating increases internal energy: the temperature rises or the state changes.
- Temperature rise: the particles gain kinetic energy.
- Change of state: the particles gain potential energy, and the temperature stays constant.
- Flat sections of a heating or cooling graph are changes of state.

COVER THE PAGE AND ANSWER

1 Define internal energy.

2 What are the two things that heating a substance can do?

3 What does a flat section on a heating graph show?

4 Which energy of the particles increases when a substance melts?

I CAN...

- state what internal energy is
- describe what heating does to the particles
- explain why the temperature is constant during a change of state
- interpret heating and cooling graphs

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–5. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

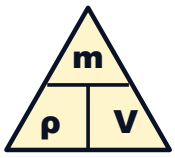
$$\rho = \frac{m}{V}$$

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$

Lesson 3



1
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

2
Write the equation in symbols.

3
Rearrange if you need to, then substitute the numbers.

4
Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS BOOKLET

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Density (ρ) L3: The mass of each unit of volume, in kg/m³ or g/cm³

Volume L3: The amount of space something takes up, in m³ or cm³

Internal energy L5: The total kinetic energy and potential energy of all the particles that make up a system

Melting point L5: The temperature at which a substance changes between solid and liquid

Boiling point L5: The temperature at which a substance boils, or condenses

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 3: DENSITY

DO NOW

1. Very close together, in a regular pattern.
2. It increases.
3. 0.250 kg

COVER THE PAGE

1. $\rho = \frac{m}{V}$
2. kg/m³ (or g/cm³)
3. 1000 kg/m³
4. 4.0 g/cm³

LESSON 5: INTERNAL ENERGY AND HEATING CURVES

DO NOW

1. $\rho = \frac{m}{V}$
2. The average kinetic energy of its molecules.
3. Freezing.

COVER THE PAGE

1. The total kinetic and potential energy of all the particles in a system.
2. Raise its temperature or change its state.
3. A change of state (melting or boiling).
4. Potential energy.